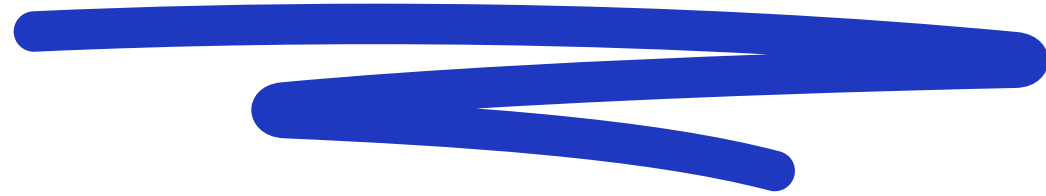


# Azure Virtual Machine with a GPU

Tutorial: how to login & use it



# 1. Email



You should receive an email with the invite to join the lab. Click the blue button.



[mad-supertomasi@unitrento365.onmicrosoft.com](#)

**ti ha invitato al lab:  
DeepLearning-Ricci**

Iscriviti subito per accedere alle macchine virtuali nel lab.

**Registrati al lab >**

Questo messaggio e-mail è stato utile? [Sì](#) [No](#)

[f](#) [X](#) [▶](#) [in](#)

[Informativa sulla privacy.](#)

Microsoft Corporation, One Microsoft Way, Redmond, WA 98052

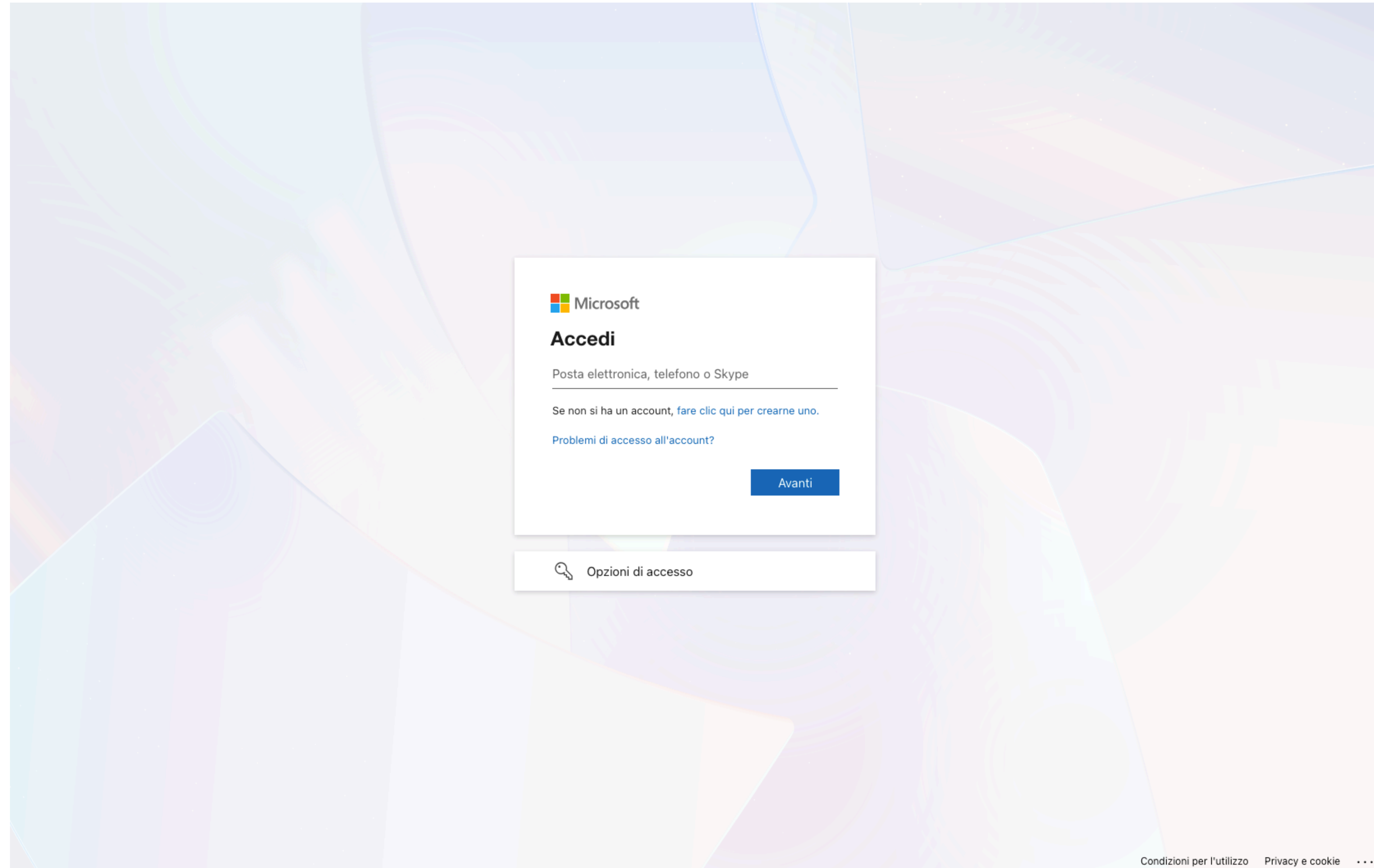


## 2. Login



You will be redirected to this login page.

**Use your @studenti.unitn.it account!** (See next slide)





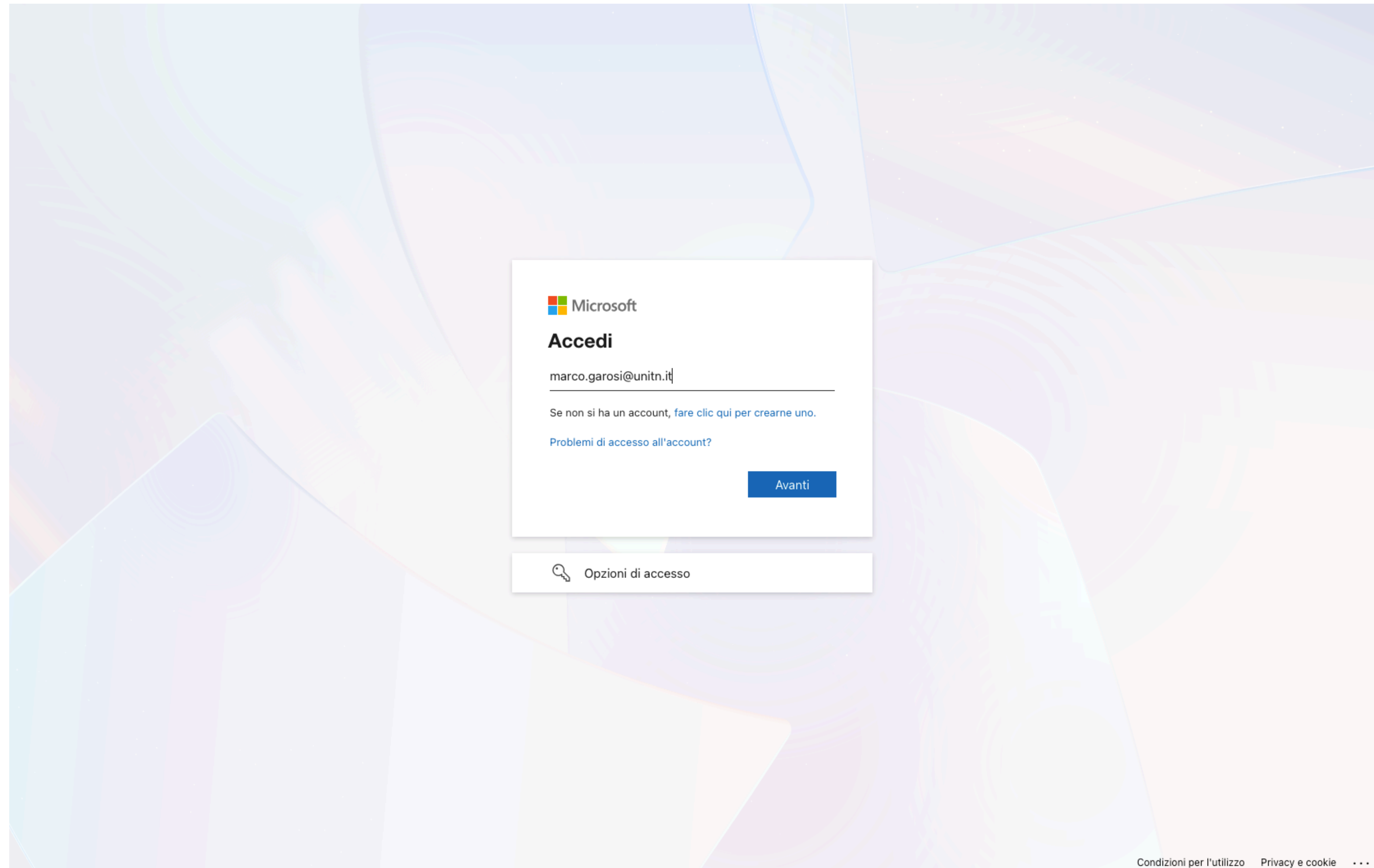
## 2. Login



**Use your @studenti.unitn.it account!**

Simply enter your email and hit the blue button.

**Note: only the student who received the email can do this!**



The screenshot shows a Microsoft login interface. At the top left is the Microsoft logo. Below it, the heading "Accedi" is displayed. A text input field contains the email address "marco.garosi@unitn.it". Below the input field, there is a link that says "Se non si ha un account, fare clic qui per crearne uno." and another link that says "Problemi di accesso all'account?". A blue button labeled "Avanti" is positioned to the right of the input field. At the bottom of the main content area, there is a link with a key icon labeled "Opzioni di accesso". The background of the page features a colorful, abstract pattern of overlapping circles and lines in shades of blue, green, and orange.

Microsoft

**Accedi**

marco.garosi@unitn.it

Se non si ha un account, [fare clic qui per crearne uno.](#)

[Problemi di accesso all'account?](#)

Avanti

 [Opzioni di accesso](#)

Condizioni per l'utilizzo Privacy e cookie ...

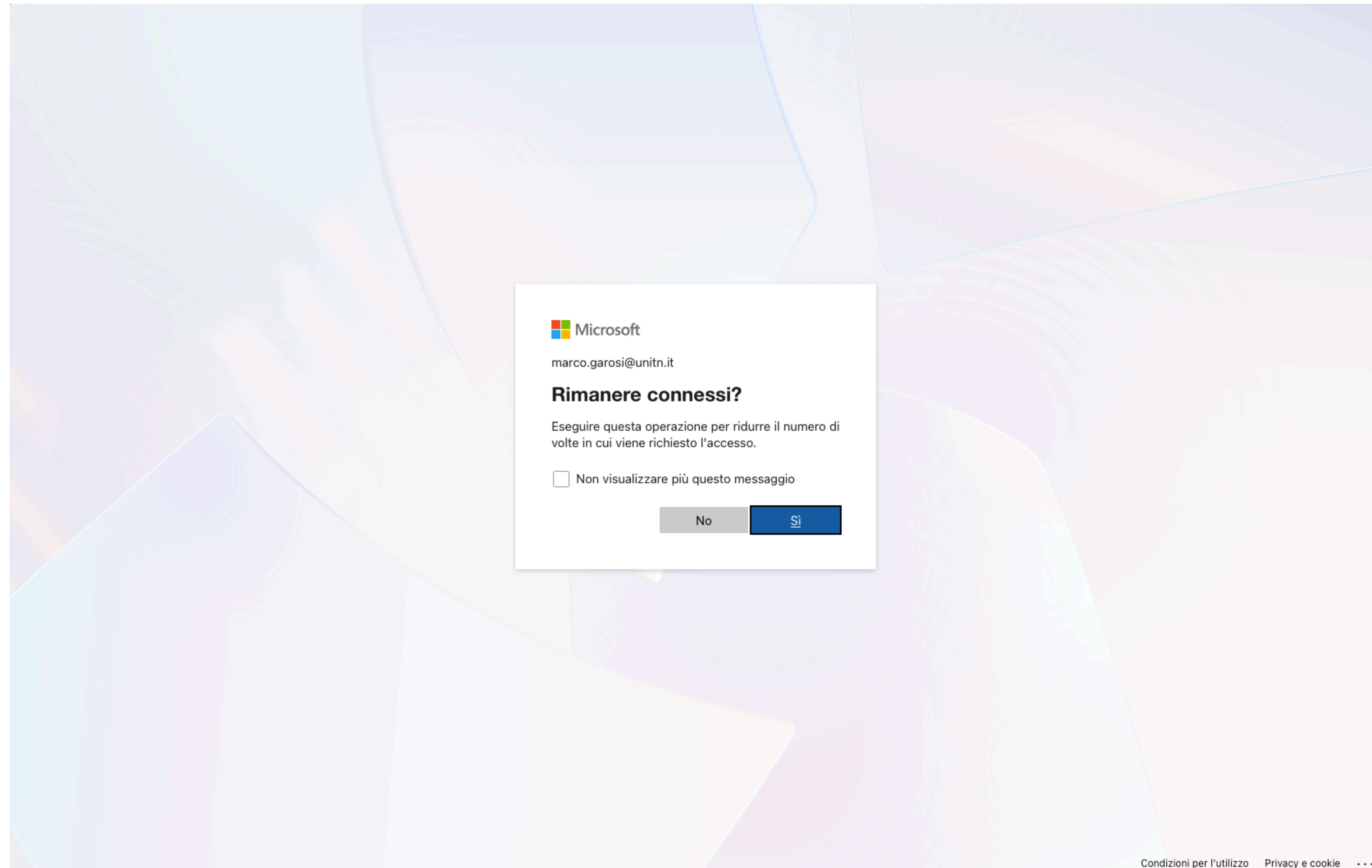
Log in using your data, and hit the red button.




## 2. Login



You will be redirected to Azure's login page. You can choose to remain connected to the account by hitting "Yes" (blue button).



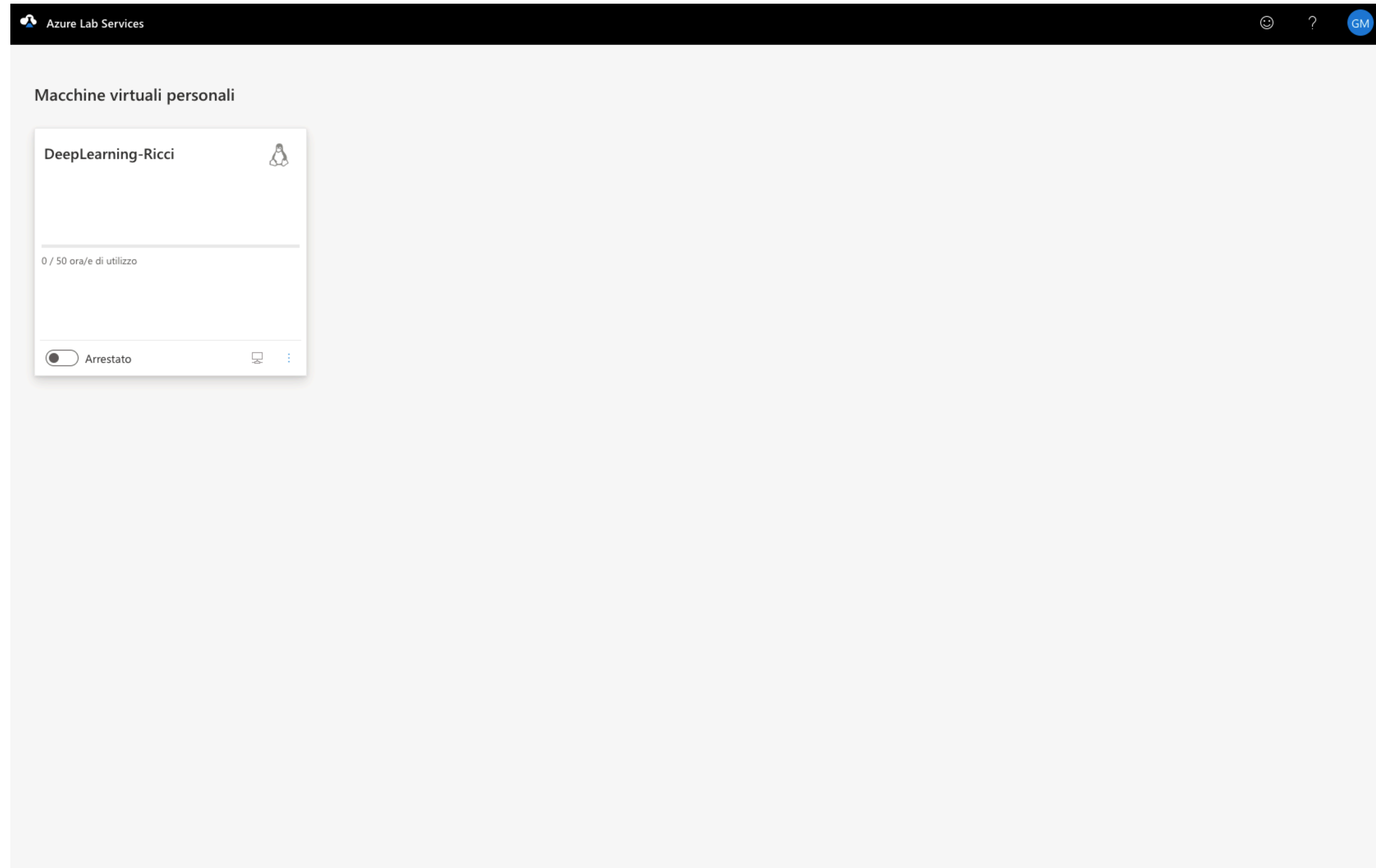


## 3. Dashboard



You are finally in the dashboard.  
You should see something like this.  
This is your virtual machine, and  
you can see how many hours you've  
used from your budget.

**Note: remember to always turn  
off the VM when you are done  
using it, otherwise you will  
waste hours!**



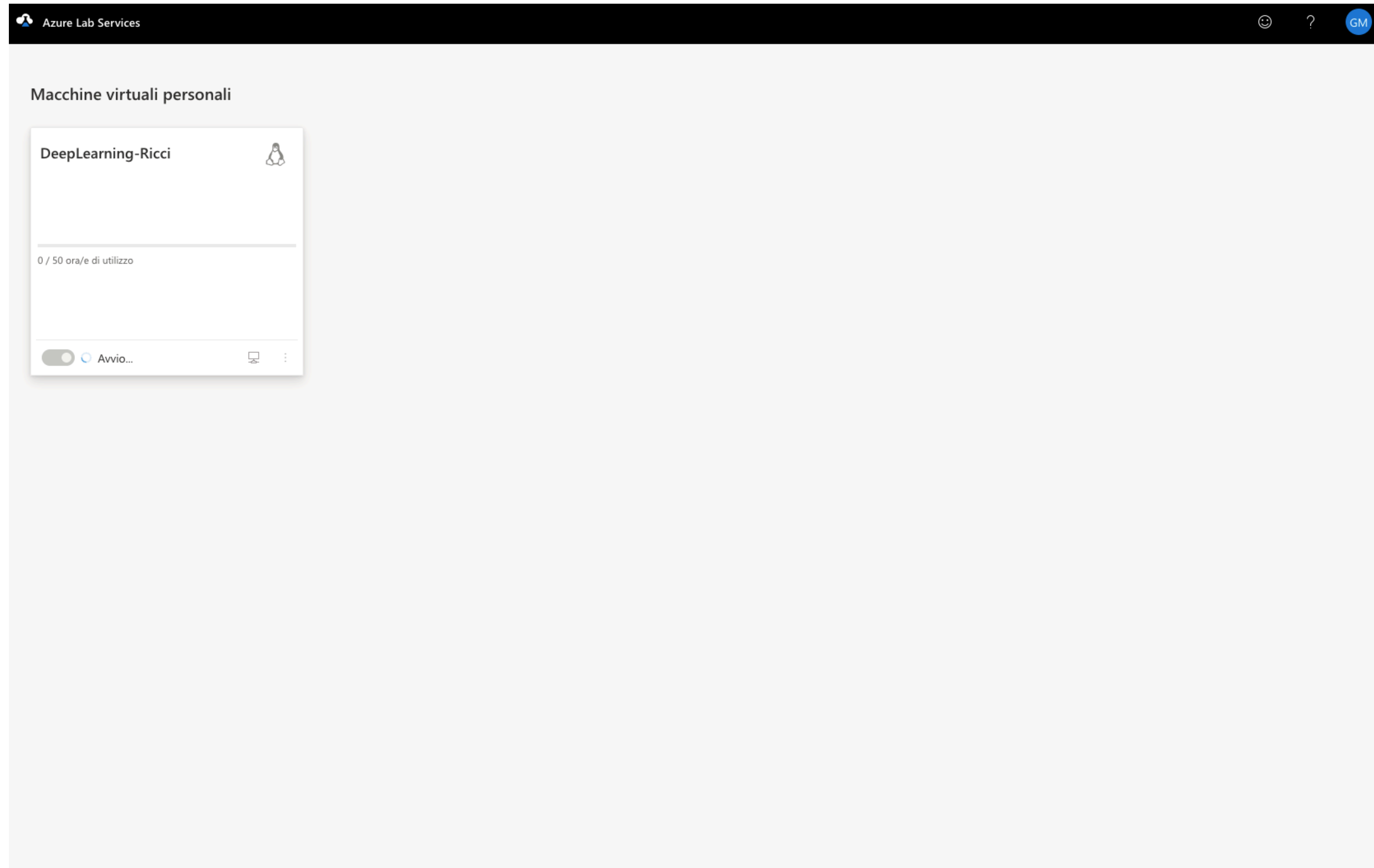


## 4. Start the VM



Click the button on the bottom-left side of the virtual machine's card. That will boot the VM.

**Note: it might takes a while to do that! Be patient and don't refresh the page.**

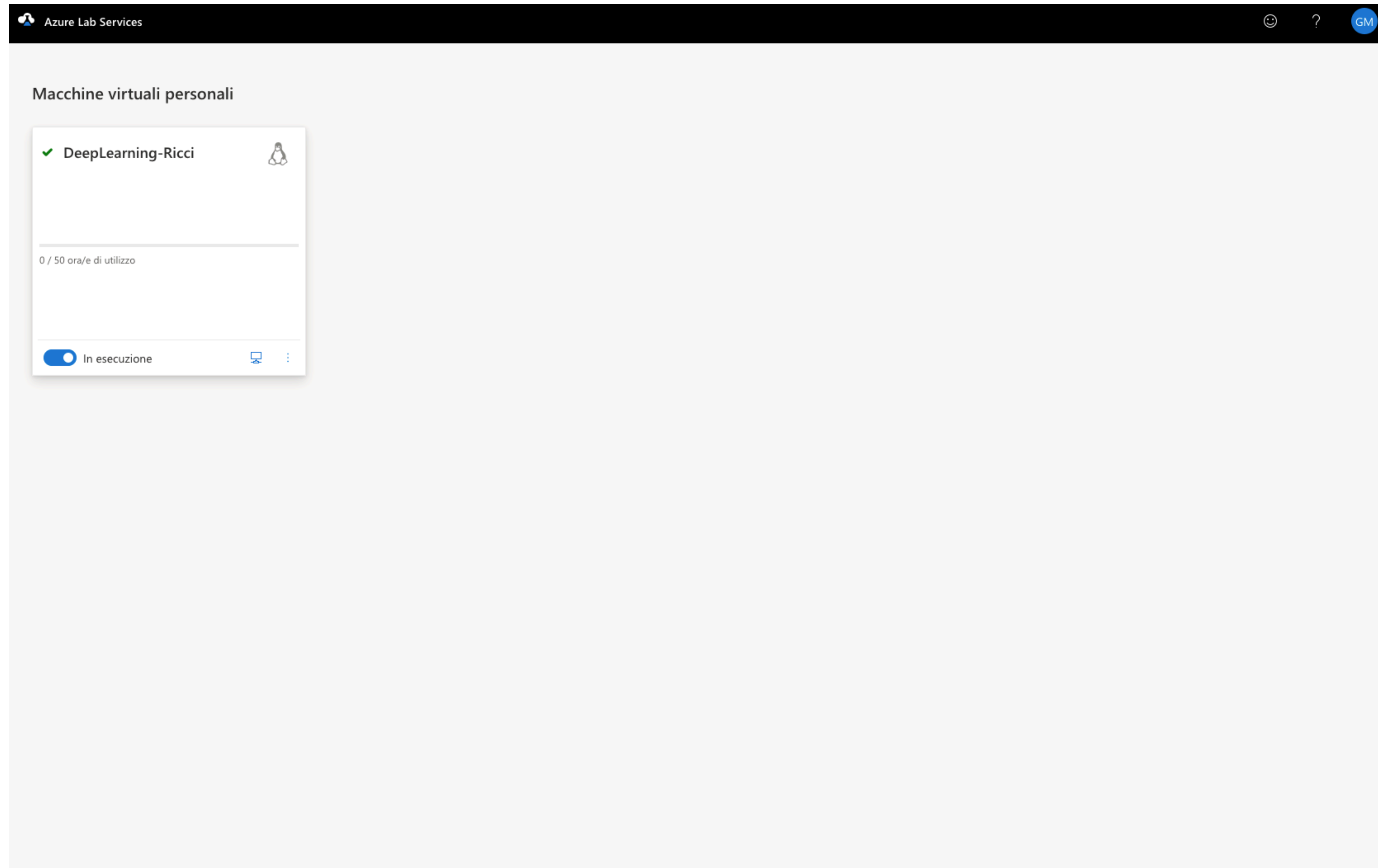




## 4. Start the VM



Once the machine is booted, the slider will say “In esecuzione” or “Running”. The VM is ready.



## 5. Set a password



You can log into the VM via SSH. However, you need to set a password. You can do so by clicking the circled icon. It will open the popup shown here.

The screenshot displays the Azure Lab Services interface. At the top, the header reads 'Azure Lab Services'. Below it, the section 'Macchine virtuali personali' (Personal virtual machines) is visible. A card for a virtual machine named 'DeepLearning-Ricci' is shown, featuring a green checkmark, a Linux logo, and a status of 'In esecuzione' (Running) with a toggle switch. A red circle highlights a small icon representing a terminal or SSH connection. A modal dialog box titled 'Imposta password' (Set Password) is open in the foreground. It contains instructions in Italian: 'Immettere una nuova password da usare per l'accesso. L'impostazione della password può richiedere qualche minuto.' (Enter a new password to use for access. Setting the password may take a few minutes). The dialog includes a 'Nome utente' (Username) field with the value 'disi' and a 'Password' field with a toggle to show or hide the password. At the bottom of the dialog are two buttons: 'Imposta password' (Set Password) and 'Annulla' (Cancel).

## 5. Set a password



Just choose a password of your choice. You can always reset it. You can share it with your teammates, so that they can also log into the VM.

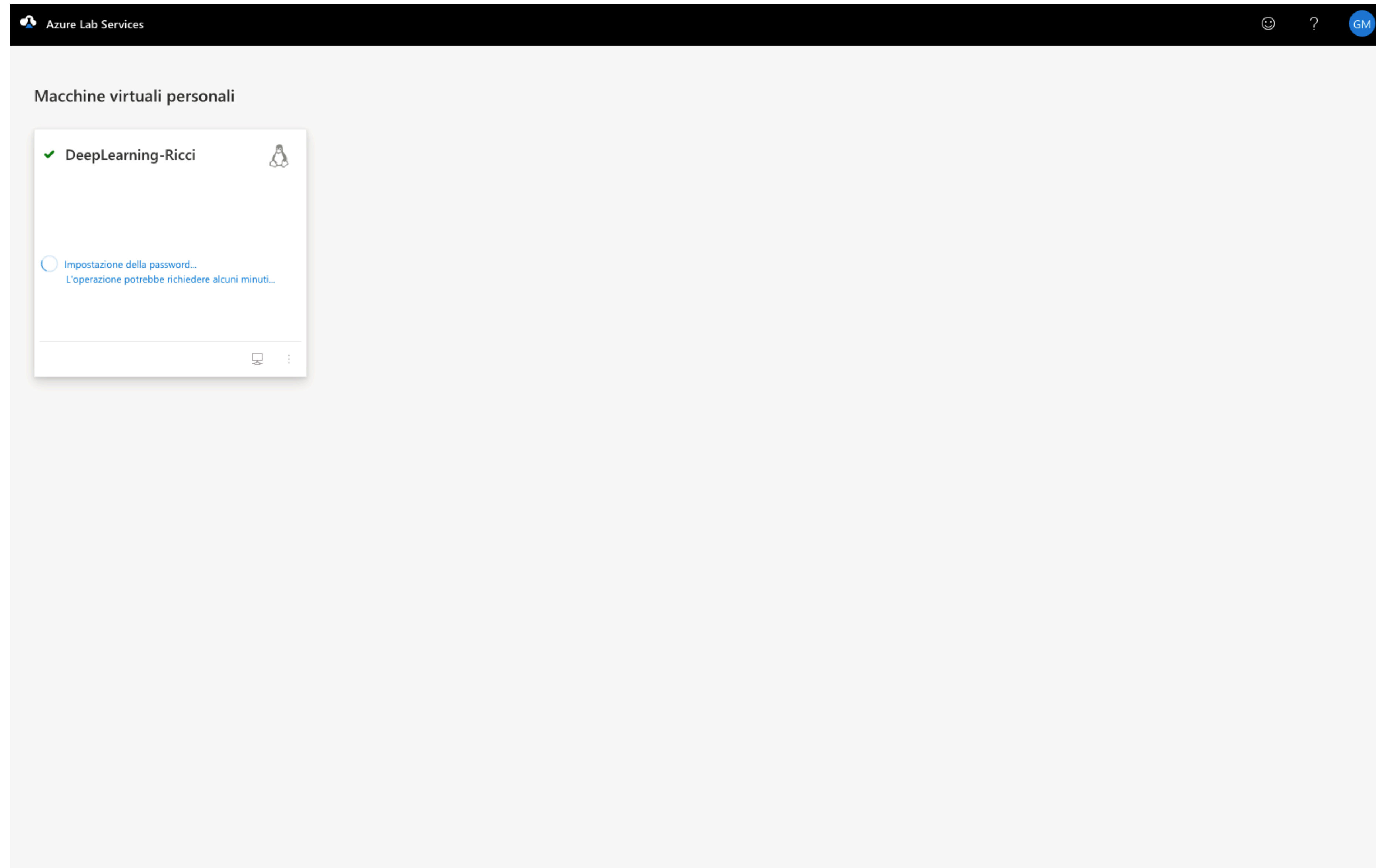
Hit the blue button when you've set the password.

**Note: if the VM is off, you can't log into it. So the "admin" will have to turn on the VM for the teammates, too.**

The screenshot shows the Azure Lab Services interface. At the top, there's a header bar with the Azure Lab Services logo and a user profile icon labeled 'GM'. Below the header, the main area is titled 'Macchine virtuali personali'. On the left, there's a card for a virtual machine named 'DeepLearning-Ricci' with a green checkmark and a Linux icon. Below the name, it shows '0 / 50 ora/e di utilizzo'. At the bottom of the card, there's a toggle switch labeled 'In esecuzione' which is turned on, and a blue button with a monitor icon. A modal dialog box titled 'Imposta password' is open in the center. It contains the following text: 'Immettere una nuova password da usare per l'accesso. L'impostazione della password può richiedere qualche minuto.' Below this, there are two input fields: 'Nome utente' with the value 'disi' and 'Password' with a masked password '.....'. At the bottom of the dialog, there are two buttons: 'Imposta password' (blue) and 'Annulla' (white).



## 5. Set a password

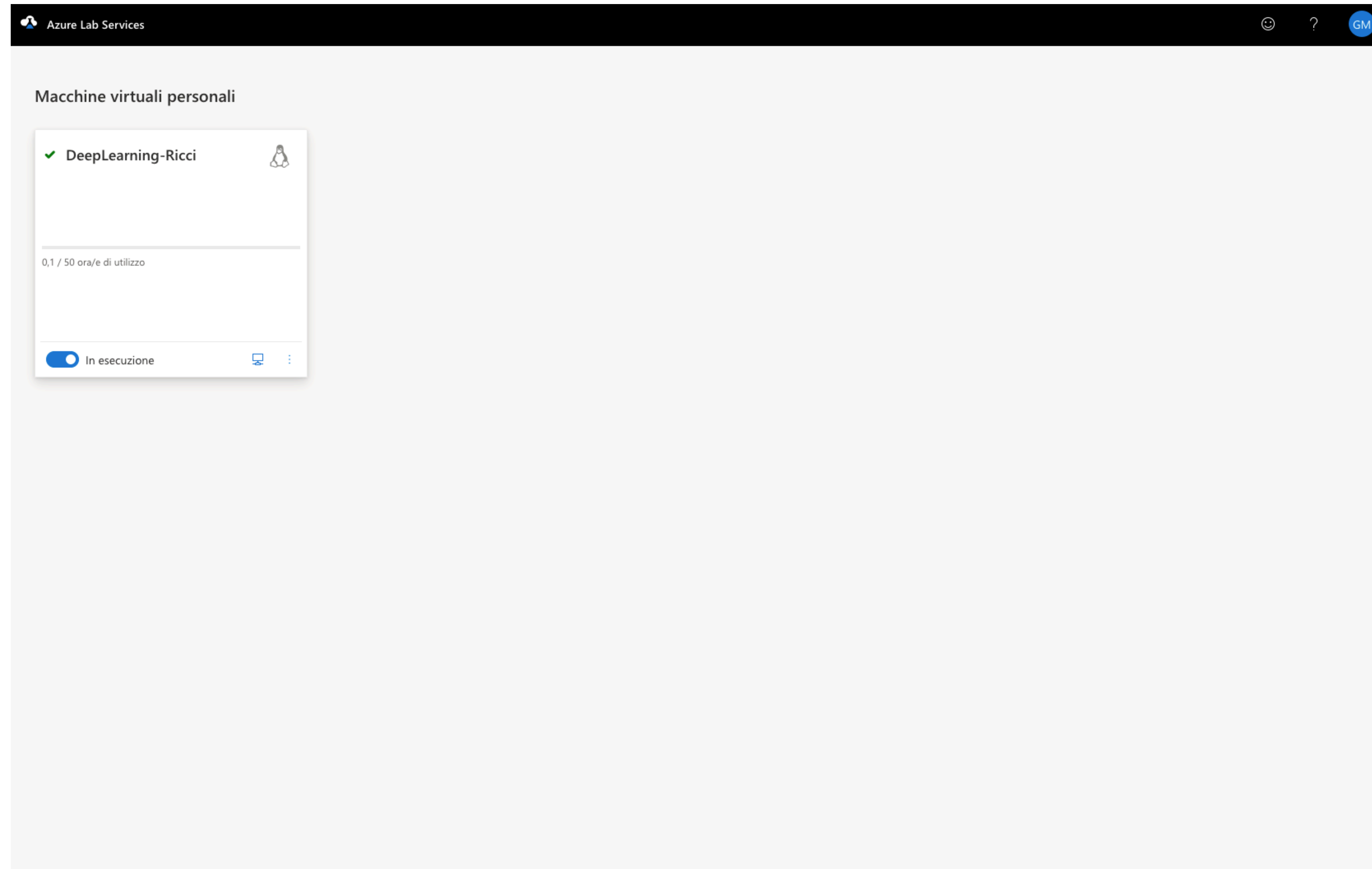


Wait a couple of minutes so that the password is correctly set. Don't refresh the page.

## 5. Set a password



Once it's done, you should see something like this. The initial setup is completed, and you can now log in into the VM.



## 6. Login



If you click again the icon in the red circle, it should now show you the address of your machine. It provides you the command to connect via SSH. **Copy-paste it.**

The screenshot displays the Azure Lab Services interface. At the top, the header reads "Azure Lab Services". Below it, the section "Macchine virtuali personali" (Personal virtual machines) is visible. A card for a virtual machine named "DeepLearning-Ricci" is shown, featuring a green checkmark, a Linux penguin icon, and a usage indicator "0,1 / 50 ora/e di utilizzo". At the bottom of the card, there is a toggle switch labeled "In esecuzione" (Running) which is turned on, and a small icon of a computer monitor with a red circle around it. A modal dialog titled "Connetti alla macchina virtuale" (Connect to the virtual machine) is open in the foreground. It provides instructions: "Per connettersi alla macchina virtuale Linux tramite SSH, usare il comando seguente:" (To connect to the Linux virtual machine via SSH, use the following command:). Below this, a text box contains the SSH command: `ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com`. There is a "Copia" (Copy) button with a clipboard icon and a "Fine" (Close) button at the bottom right of the modal.



## 6. Login



Move to your terminal  
and paste the command.  
Then, hit Enter.

```
➤ ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
```



## 6. Login



If something like this happens, just type in “yes” and hit Enter.

```
➤ ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:1Ln8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? █
```

## 6. Login



Type in your password (the one you set earlier in Azure's web console) and hit Enter.

**Note: the password will not appear, and you will not see any blinking indicator (as it usually happens in the terminal). Just type the password and hit Enter, your keyboard is working!**

```
➤ ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:1Ln8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040' (ED25519) to the list of known hosts.
disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com's password: ʔ
```

## 6. Login



If you did everything correctly, you should see something like this. You have eventually made it to the VM!

You are an admin to this VM, so you can do (almost) whatever you want with it.

You can test the GPU with `nvidia-smi`, which will show you some info about the GPU.

```
➤ ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:lLn8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040' (ED25519) to the list of known hosts.
disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Apr 23 10:56:10 UTC 2026

System load:  0.08               Processes:            166
Usage of /:   14.3% of 122.94GB   Users logged in:     0
Memory usage: 0%                IPv4 address for eth0: 10.0.0.44
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

18 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

disi@labVDMJXW:~$
```



## 6. Login

```
disi@labT31YWB:~$ nvidia-smi
Thu Apr 23 12:19:13 2026
+-----+
| NVIDIA-SMI 555.42.06                  Driver Version: 555.42.06          CUDA Version: 12.5        |
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan   Temp   Perf              Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|                               |                      |              MIG M. |
+-----+-----+
|  0   Tesla V100-PCIE-16GB                Off | 000000001:00:00.0 Off |                    |
| N/A    30C    P0              25W / 250W |      1MiB / 16384MiB |      0%      Default |
|                               |                      |              N/A    |
+-----+-----+

+-----+
| Processes:                               |
|  GPU   GI    CI          PID    Type   Process name                  GPU Memory |
|          ID    ID                                   Usage                 |
+-----+-----+
| No running processes found              |
+-----+
```

`nvidia-smi` shows something like this:



## 7. Environment setup



Now you need to install Python and PyTorch.  
Using uv will make this easy.

Just type in the following commands:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
source $HOME/.local/bin/env
mkdir <your_project_folder>
cd <your_project_folder>
uv init --python 3.12
uv venv
source .venv/bin/activate
uv pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
```



## 7. Environment setup



This is the output that you should expect from the commands in the previous slide.

```
disi@labT31YWB:~$ curl -Lsf https://astral.sh/uv/install.sh | sh
downloading uv 0.11.7 x86_64-unknown-linux-gnu
installing to /home/disi/.local/bin
  uv
  uvx
everything's installed!

To add $HOME/.local/bin to your PATH, either restart your shell or run:

    source $HOME/.local/bin/env (sh, bash, zsh)
    source $HOME/.local/bin/env.fish (fish)
disi@labT31YWB:~$ source $HOME/.local/bin/env
disi@labT31YWB:~$ mkdir test
disi@labT31YWB:~$ cd test/
disi@labT31YWB:~/test$ uv init --python 3.12
Initialized project `test`
disi@labT31YWB:~/test$ uv venv
Using CPython 3.12.13
Creating virtual environment at: .venv
Activate with: source .venv/bin/activate
disi@labT31YWB:~/test$ source .venv/bin/activate
(test) disi@labT31YWB:~/test$ uv pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
Resolved 32 packages in 1.64s
Prepared 32 packages in 27.91s
Installed 32 packages in 264ms
+ cuda-bindings==12.9.4
+ cuda-pathfinder==1.2.2
+ cuda-toolkit==12.6.3
+ filelock==3.25.2
+ fsspec==2026.2.0
+ jinja2==3.1.6
+ markupsafe==3.0.3
+ mpmath==1.3.0
+ networkx==3.6.1
+ numpy==2.4.3
+ nvidia-cublas-cu12==12.6.4.1
+ nvidia-cuda-cupti-cu12==12.6.80
+ nvidia-cuda-nvrtc-cu12==12.6.85
+ nvidia-cuda-runtime-cu12==12.6.77
+ nvidia-cudnn-cu12==9.10.2.21
+ nvidia-cufft-cu12==11.3.0.4
+ nvidia-cufile-cu12==1.11.1.6
+ nvidia-curand-cu12==10.3.7.77
+ nvidia-cusolver-cu12==11.7.1.2
+ nvidia-cusparselt-cu12==0.7.1
+ nvidia-cusparselt-cu12==0.7.1
+ nvidia-nccl-cu12==2.28.9
+ nvidia-nvjitlink-cu12==12.6.85
+ nvidia-nvshmem-cu12==3.4.5
+ nvidia-nvtx-cu12==12.6.77
+ pillow==12.1.1
+ setuptools==70.2.0
+ sympy==1.14.0
+ torch==2.11.0+cu126
+ torchvision==0.26.0+cu126
+ triton==3.6.0
+ typing-extensions==4.15.0
```

## 8. Test the installation



Now you can test the GPU  
(and if PyTorch recognizes it)  
by running Python in  
interactive mode, as shown  
here.

No errors: everything is  
working fine.

```
(test) disi@labT31YWB:~/test$ python
Python 3.12.13 (main, Apr 14 2026, 14:29:00) [Clang 22.1.3 ] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> torch.cuda.is_available()
True
>>> x = torch.zeros((100, 100), device="cuda")
>>> x.device
device(type='cuda', index=0)
>>> x ** 2
tensor([[0., 0., 0., ..., 0., 0., 0.],
        [0., 0., 0., ..., 0., 0., 0.],
        [0., 0., 0., ..., 0., 0., 0.],
        ...,
        [0., 0., 0., ..., 0., 0., 0.],
        [0., 0., 0., ..., 0., 0., 0.],
        [0., 0., 0., ..., 0., 0., 0.]], device='cuda:0')
>>> exit()
```



# Turn off

Just a final reminder to turn off the VM when you're not using it, otherwise you will run out of budget pretty quickly.

# the VM!

